100 Days of Python

Day 54 — Practice Assignment: List Comprehension Problems

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1. **Given words = ["python", "is", "fun", "and", "powerful"], use list comprehensions to build a list of the length of each word, and a second list of only the words longer than 3 letters.**

   Hint: the first keeps every word and puts len(w) in the list; the second puts w in the list, with an if at the end.

2. **Build a list of every multiple of 3 from 1 to 30 using one list comprehension.**

   Hint: n % 3 == 0 is True when n divides evenly by 3. Remember range(1, 31) to include 30.

3. **Given temperatures in Celsius [0, 100, 37], build a list of the same temperatures in Fahrenheit using the formula c * 9 / 5 + 32.**

   Hint: every temperature is kept, so there is no if at all.

4. **Give each mark in [45, 95, 89] a bonus of 10, but make sure no mark ends up above 100. Print the new list.**

   Hint: min(m + 10, 100) picks the smaller of the two values.

5. **From fruits = ["apple", "banana", "orange", "kiwi", "egg"], build a list holding only the fruits that start with a vowel.**

   Hint: word[0] is the first letter, and "aeiou" can be used with in to test it.

6. **Given marks = [45, 67, 34, 90, 52], print how many marks are passes (40 or above), and then the average of just those passing marks.**

   Hint: build the list of passing marks once, then use len() and sum() on it.

7. **Challenge: given words = ["python", "is", "fun", "and", "powerful"], print the different word lengths, each listed once, from smallest to largest.**

   Hint: build the list of lengths with a comprehension, then wrap it in set() and sorted().